



How do incumbents use AI technologies to strategically renew their innovation portfolios?

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ABSTRACT

The strategic use of Artificial Intelligence (AI) to enhance firms' innovation and drive sector-specific technological progress remains underexplored. This study proposes a typology of AI-based inventions, distinguishing between *Discontinuous Strategic Renewal Inventions (DSRI)* that initiate new technological directions, *Follow-on DSRI* that extend developments along those new paths, and *Continuous Strategic Renewal Inventions (CSRI)* that augment existing innovation portfolios. Focusing on the agricultural sector, we analyse a dataset of 14,400 patents from incumbents and new entrants that have leveraged their expertise to integrate AI technologies into their innovation strategies. Using negative binomial regression, we investigate the impact of these AI invention types on strategic renewal efforts. Our results underscore the pivotal role of *DSRI*, particularly *Follow-on DSRI*, in driving technological advancement. Contrary to prevailing assumptions, we find that the benefits of strategically implementing AI vary markedly between industries, with traditional agricultural firms typically making greater gains than entrants from the electronics or machinery sectors. The study underscores the strategic imperative for old and new players in the market to persistently explore disruptive AI technologies to unlock new technological avenues and competitive advantages.

1. Introduction

Artificial Intelligence (AI) has led to fundamental changes in the ways in which companies develop new products (Verganti et al., 2020; Gregory et al., 2021; Tong et al., 2021), and create and capture value (Bradley and O'Toole, 2016; Sestino et al., 2023), blurring traditional company boundaries along value chains in various markets (Appio et al., 2021; Marion and Fixson, 2021; Majumdar et al., 2018). While AI has opened doors for new entrants, it is also leveraged by incumbents¹ to renew their innovation portfolios, maintain the status quo, and generate new revenue streams, even outside of their core areas of expertise (Cockburn et al., 2018; Verganti et al., 2020; Gregory et al., 2021).

AI impacts incumbents in different industries in a variety of different ways (Felten et al., 2021; Plantec et al., 2023), which often depend on

how they embrace the technology (Ransbotham et al., 2020). The new business and organisational opportunities include the use of AI to increase production efficiency (Cockburn et al., 2018; Tong et al., 2021), or as a disruptive force to extend their own and their market's innovation portfolios (Appio et al., 2021; Verganti et al., 2020; Rana et al., 2017; Sestino et al., 2023; Sestino and de Mauro, 2022). However, these opportunities not only depend on industry specifics, such as a highly turbulent or relatively stable environment, but also on a general capacity to adapt and survive (Raj and Seamans, 2019), given that responses to technological changes are largely heterogeneous (Eggers and Park, 2018). Moreover, it has been shown that in the face of major technological disruptions, such as those triggered by AI, incumbents are particularly prone to struggle to develop new product functionalities.

Certain gaps in the literature limit our understanding of how

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¹ Companies with an established presence and typically high market share in a given industry/sector.

incumbents extend or complement AI in their existing innovation portfolios (Ceipek et al., 2021a; Sestino et al., 2020; Ceipek et al., 2021b; Sun et al., 2020), and there have been numerous calls for further investigation in this regard (Johnson et al., 2017; Raisch and Krakowski, 2021; Plantec et al., 2023). While the literature has demonstrated how companies use AI to increase efficiency and even develop new products (Lanzolla et al., 2021; Iansiti and Lakhani, 2020), there is a dearth of knowledge on whether and how they actively deploy the technology to implement new functionalities within existing or new products, either to address emerging needs or to gain entry into new market segments (Verganti et al., 2020). It is critical for incumbents to consider ‘aviansation’, i.e. their capacity to strategically renew their resources and remain competitive in their ‘playing field’² (Roy et al., 2018; Agarwal and Helfat, 2009; Schmitt et al., 2018; Tuncdogan et al., 2019). In this paper, we describe anything that falls within the technological boundaries of a given playing field as ‘local’ (Capron and Mitchell, 2009). Hence, ‘local companies’ are incumbents or small and medium-sized enterprises that focus their main activities within a specific playing field, while ‘external companies’ are new entrants seeking to capitalise on business opportunities or apply their technologies in that same field.

By addressing these gaps in the literature, we are able to improve the general understanding of how companies should navigate the challenges of integrating AI into new products and, more generally, how digital technologies may help resolve the issue of path-dependent behaviour (Bohnsack et al., 2021).

Drawing on the strategic renewal literature, we explore (1) the various ways in which incumbents within a specific playing field integrate AI technologies into their innovation portfolios, and (2) the impact of these different modes of integration on future technological developments within the same playing field.

These are critical questions, for despite the burgeoning interest in AI’s potential, the strategic renewal literature has yet to fully explain how incumbents in specific industries integrate it into their innovation portfolios. Moreover, the implications of such integration for future technological development remain underexplored. This gap is particularly pronounced with regard to how AI catalyses different strategic renewal processes and shapes innovation trajectories across industries. Existing studies have primarily focused on AI adoption in a general sense, overlooking its nuanced effects on incumbents’ renewal practices and the subsequent technological evolution within specific fields.

To address these gaps, we analyse one of the oldest industries of all, namely agriculture. The AI-driven transformation of farming is just the latest of many technological disruptions dating back to the dawn of human civilisation, including Neolithic stone tools, wind-powered irrigation, and many other examples. Applications of AI to agriculture include automation (Jha et al., 2019), decision support (Smith, 2020), crop yield prediction (Dahikar and Rode, 2014), optimising pest and disease management (Bannerjee et al., 2018), and addressing some of the major environmental challenges in the sector, such as climate change, plant health, biodiversity loss and feeding a growing global population (Ryan et al., 2018; Vinuesa et al., 2020; Acatech and Reisch, 2020; National Academies of Sciences, Engineering, and Medicine, 2019; FAO, 2009; Rennings et al., 2024).

The agricultural sector features both long-standing incumbents like DEERE & Co, which revolutionised farming by introducing the steel plough in 1833 and is now developing self-driving tractors (Knight, 2022), and new entrants such as IBM, which is advancing early-stage AI integration. Not only is this a useful sector for addressing our research questions, but it will also be interesting to observe how AI contributes to broader efforts to transform agriculture toward greater sustainability. We should note, however, our focus here is solely on strategic renewal

driven by new developments in AI technologies. Other forms of strategic renewal, such as new business models, fall out of the scope of this paper and warrant separate research elsewhere.

We begin by developing a typology that distinguishes between three different types of AI-based inventions from a strategic renewal perspective (Agarwal and Helfat, 2009; Crossan and Berdrow, 2003). Those that open new technological paths for the company are called *Discontinuous Strategic Renewal Inventions (DSRI)*. Those that build on these paths are called *Follow-on DSRI* and those that use AI to expand the existing technology portfolio without disrupting the core business are labelled *Continuous Strategic Renewal Inventions (CSRI)*.

Second, we explore how what we have defined as local and external companies in the specific case of the agricultural playing field harness these inventions to improve or extend their existing innovation portfolios. Using negative binomial regression, we evaluate the impact of each invention type on future technological developments.

By capitalizing on our typology and patent-based analyses, we show that *DSRI*, especially *Follow-on DSRI*, have a more substantial impact on future technological developments than *CSRI*. In other words, strategies that disrupt the existing business model through AI integration are more impactful than those that merely extend the current portfolio. Furthermore, the value derived from AI-based strategic renewal is contingent on a firm’s main industry of origin. Particularly in the agricultural sector, incumbents benefit more from *DSRI* than companies entering from the machinery or electronics sectors. Hence, an effective innovation framework needs to balance risk-taking against sustained commitment to AI integration.

2. Theoretical background: AI-based inventions for strategic renewal

Strategic renewal is the refreshment or replacement of organisational attributes that determine a firm’s long-term growth and survival in dynamic environments (Agarwal and Helfat, 2009; Roy et al., 2018; Schmitt et al., 2018; Helfat and Winter, 2011). It has been defined as “the activities a firm undertakes to alter its path dependence” (Volberda et al., 2001, p. 160) and as a transformation of capabilities to maintain competitive advantage (Schmitt et al., 2018). Companies adopt distinct practices to achieve this (Kwee et al., 2011). For example, Agarwal and Helfat (2009) argue that companies pursue such goals through both discontinuous transformation and incremental renewal. The latter includes proactive adaptations such as experimentation with AI-driven business models in the early stages or incremental alterations to core products or other inventions through corporate venturing.

2.1. Typology of AI-based inventions for strategic renewal

AI technologies have been described as “learning system(s); that is, machines that can become better at a task typically performed by humans with limited or no human intervention” (WIPO, 2019b, p. 19). As a discipline of computer sciences, AI involves designing systems that have some kind of ‘intelligence’ (Patterson, 1990), such as learning, reasoning, and understanding language or visual cues; functions traditionally associated with human intelligence (Russell and Norvig, 2003).

Owing to the wide range of tasks that AI can perform, they are central to digitalisation efforts in modern economies, and can be cheaper, faster, and more reliable than humans (Castelli et al., 2016; Manyika, 2017). With their enormous potential to increase the efficiency of the global economy (Cockburn et al., 2018), AI is expected to be one of the most disruptive technologies in the coming decades (Rana et al., 2017), particularly with regard to new product development (Verganti et al., 2020). Often viewed as an enabling or general-purpose technology, it has the potential to affect almost every industry (Klinger et al., 2018; Bresnahan and Trajtenberg, 1995; Brynjolfsson and McAfee, 2017), to the extent that it now being dubbed the fourth industrial revolution (Schwab and Davis, 2018). However, AI is still in its infancy and requires

² The playing field refers to the general conditions or framework, such as a market or industry/sector, that shape an actor’s behaviour (Beck and Demircik-Kunt, 2006).

“additional innovation, both in terms of technology directly and in terms of the development of new business practices” (Goldfarb et al., 2020, p. 1), underscoring the need for additional research on how firms across industries effectively use AI to drive innovation and achieve technology-based strategic renewal.

AI technologies are changing companies' management practices (Schneider and Leyer, 2019), innovation processes (Cockburn et al., 2018), innovation teams (Bouschery et al., 2023), businesses strategies (Iansiti and Lakhani, 2020), and ways of creating and capturing value (Bradley and O'Toole, 2016) — with current developments around transformer-based models such as ChatGPT further fuelling the debate (see e.g., Bouschery et al., 2023). However, companies differ in their reliance on AI (Cennamo et al., 2020), with Lanzolla et al. (2021) noting that the ways in which they combine it with external knowledge depends largely on strategic and managerial choices. While AI integration is essential for developing successful new products, effective implementation is often constrained by low adoption rates or insufficient human resources (Boland et al., 2007; Vuori and Huy, 2016), hence the interest in an in-depth understanding of how AI technologies are implemented in incumbents' innovation portfolios.

We begin our inquiry into the matter by exploring pre-existing concepts in the AI literature on computer sciences, where the difference between weak and strong AI has not only been addressed in the academic literature but also in the press (e.g., Walch, 2019; Frankish and Ramsey, 2014). As early as 1980, Searle was already making this distinction while reflecting on the psychological and philosophical significance of attempts to get computers to ‘think’ in a similar manner to a human brain: “[In strong AI] the appropriately programmed computer really is a mind, in the sense that computers given the right programmes can be literally said to understand and have other cognitive states” (Searle, 1980, p. 417). But at least for now, AI does not fully act like a human mind, and strong AI is more a matter for philosophers than a business reality. It is weak AI that does exist and which can lead to significant changes in our modern economies (Iansiti and Lakhani, 2020). In management science, this variant has been defined as “computer systems [which] perform tasks that were traditionally performed by human beings” (Iansiti and Lakhani, 2020, p. 8). Indeed, robots or algorithms are able to reproduce routine human tasks, such as processing data or recognising images, faster, more reliably, and often at a lower cost, with clear implications for productivity and production (Levy, 2018; Jain et al., 2017).

Weak AI-based inventions do not always replace human labour directly. They can also create entirely new business models or tasks without having a ‘mind’ per se. Hence, other inventions need to be conceptualised in which AI technologies such as robotics and deep learning are leveraged to design artefacts that perform entirely new tasks beyond the dominant innovation portfolio of a given market or company. These inventions allow companies to overcome cognitive lock-in, the tendency to frame functions or artefacts solely within traditional uses (Kaplan and Tripsas, 2008; Thrane et al., 2010) and instead produce novel solutions. In the agricultural sector, examples include automated sensor-based weeding systems, using unmanned aerial vehicles (UAVs, or drones) to identify low-yield areas, pests, or diseases (Jarman et al., 2016), edge-of-field monitoring devices to assess the quality of runoff water (Harmel et al., 2018), data visualization software to monitor farm operations (OECD, 2019), data-driven systems to track harvest quality (Jouanjan, 2019), and the use of remote sensing for crop monitoring. Not only can these innovations significantly reduce manual labour — still, the incorporation of human expertise often proves indispensable in agriculture, leading to human-centred AI approaches (see e.g. Holzinger et al., 2024; Holzinger et al., 2022) — and improve crop quality, but they have also given rise to completely new business activities (OECD, 2019). We call these *Discontinuous Strategic Renewal Inventions (DSRI)*, and define them as follows:

Discontinuous Strategic Renewal Invention (DSRI): An invention that relies on weak AI technologies to perform functions in different

domains to a company's previous inventions.

We note that while a given invention can provide new functionalities, it will probably require further development before it can be implemented or commercialized. Therefore, we define:

Follow-on Discontinuous Strategic Renewal Invention (Follow-on DSRI): An invention that complements pre-existing *Discontinuous Strategic Renewal Inventions*.

Here, our focus is on the novelty and competitive characteristics of the invention, rather than on the process of mobilising knowledge to design it. However, we also define the use of AI to extend a company's existing innovation portfolio without disrupting its core business, and we call this *Continuous Strategic Renewal Inventions (CSRI)*, defined as follows:

Continuous Strategic Renewal Invention (CSRI): An invention that relies on AI technologies and performs functions in domains that are similar to those of a company's previous inventions.

This typology of AI-based inventions can be used to distinguish between the different approaches that incumbents may use when integrating AI in their inventive processes, as opposed to non-AI-based inventions. We can draw on these to discuss the implications for companies' capabilities for pursuing different strategic renewal strategies and to evaluate the impact of each AI-based invention type on future technological developments within the playing field, considering both local and external companies.

2.2. Effects of developing AI-based inventions on incumbents' innovation portfolios

Incremental renewal, echoing our concept of *CSRI*, can help to routinise organisational behaviour (Martens et al., 2012). On the other hand, more profound transformation, as characterised by *DSRI*, often occurs when incumbents face declining profits, market saturation or major shifts in technology or consumer behaviour. In order to access more distant knowledge, incumbents often rely more extensively on explorative innovation (Crossan and Berdrow, 2003; Fleming, 2001), leveraging AI technologies to create new business opportunities. Once they enter new fields, acceleration is likely to occur (Blomkvist et al., 2010), as reflected by our concept of *Follow-on DSRI*.

These patterns can be contrasted against key innovation theories, such as disruptive innovation, which posits that companies may introduce simpler, more affordable products that challenge incumbents by targeting underserved or overlooked segments (Christensen and Overdorf, 2000). As with *DSRI*, disruptive innovation entails transformations that fundamentally alter the competitive landscape and demand adaptive responses. However, our typology extends beyond market disruptions alone, focusing instead on the mechanisms and strategic pathways made possible by AI technologies, whereby firms are able to redefine their innovation portfolios and enter entirely new business domains.

A similar distinction can be made between autonomous and systemic innovation (Chesbrough and Teece, 2009). Autonomous innovations are typically standalone advancements that require minimal external coordination, echoing the incremental logic of *CSRI*. In contrast, systemic innovations require integration across technologies and ecosystems, aligning more closely with the exploratory and transformative nature of *DSRI*. However, our typology goes further by emphasising the iterative nature of strategic renewal, particularly the acceleration observed in *Follow-on DSRI*.

The implementation of new technologies, and the resulting interactions among multiple actors, shapes how innovations evolve within a given playing field (Kaplan and Tripsas, 2008). While some innovations appear radical in a certain context, many had previously been incrementally developed in others (Tripsas, 2008). Incumbents (i.e., local companies) are often challenged by new entrants (i.e., external companies) that already operate in other playing fields and are looking to diversify their activities into new ones. Incumbents frequently struggle to implement disruptive innovations through discontinuous

(and follow-on) renewal strategies, and instead merely build on their existing technological foundations without connecting them to evolving innovations (Bergek et al., 2013). As a result, incumbents focus their attention on potentially path-breaking innovations and their discontinuous renewal efforts tend to differ from those of new entrants (Thrane et al., 2010). Local companies are also particularly prone to exploit their existing routines and resources to defend their position (Flier et al., 2003).

Our exploration also resonates with the Information Systems literature on strategic IT-business alignment, particularly through the lens of AI implementation and its integration into firm-level innovation processes (e.g., Venkatraman et al., 1993; Maes et al., 2000; Luftman et al., 2017 (for a review)). Strategic IT-business alignment, defined as “the coordinating activities across IT and non-IT domains within the firm in ways that are likely to provide new services, improve business processes and decision making and thereby increase the business value of the firm” (Luftman et al., 2017, p. 27), underscores the importance of linking technological initiatives to broader organisational objectives. Prior research highlights critical enablers for such alignment, including robust IT governance (Broadbent and Kitzis, 2005; Salgado-Criado et al., 2024 (in the context of venture capital)) and the strategic commitment of leadership, particularly CEOs, to IT initiatives (Hussin et al., 2002). Building on these insights, our typology of AI-based inventions addresses how firms can strategically integrate AI technologies to achieve not only immediate operational improvements but also long-term strategic renewal. This alignment is particularly critical in the context of digital transformation, where AI can foster disruptive innovation and reshape value propositions (Huang et al., 2017), provided the necessary operational and managerial conditions – such as data preparation and managerial commitment – are in place (Plantec et al., 2023). However, digital transformation efforts often encounter resistance and organisational inertia (Vial, 2021), raising key questions about the strategic use of AI for automation – specifically, which tasks to automate and at what level (Dwivedi et al., 2021). To fully harness the potential of next-generation AI for innovation, companies must align their competitive and cognitive strategies, thus enabling the creation of innovative products and solutions while overcoming barriers to adoption (Borges et al., 2021; Ku and Chen, 2024).

Finally, as the literature recommended a deeper exploration of how diverse incumbents integrate digital technologies such as AI into their innovation portfolios for different purposes (see e.g., Bohnsack et al., 2021; Bohnsack et al., 2024), our conceptualization of different AI-based inventions makes a valuable contribution to this line of research. Particularly due to the adoption of a strategic renewal perspective, the context of innovation in the agricultural sector, with the presence of both local and external incumbents, is especially relevant to this research field.

3. Method

3.1. Research focus: strategic renewal and innovation in the agricultural sector

The agricultural sector is a significant playing field in the global economy (Bronson and Knezevic, 2016; Lowenberg-DeBoer and Erickson, 2019; Rennings et al., 2023; Misra et al., 2022) and is one in which incremental and radical innovations are nothing new. After such revolutions as mechanisation and genetic enhancements, the agricultural sector is currently undergoing major strategic renewal and innovation due to the adoption of AI technologies to enhance existing operations and processes (Fountas et al., 2015). For instance, DEERE & CO has applied AI-powered predictive analytics to forecast crop yields (McBratney et al., 2005), which has significantly increased efficiency and productivity, enabling incumbents to maintain their competitive advantage.

Conversely, AI is also fostering discontinuous, or disruptive, strategic

renewal within agriculture. For example, AGCO Corp., a worldwide leader in the design, manufacture, and distribution of agricultural equipment, has embraced precision agriculture, combining AI and Internet of Things (IoT) technologies for real-time crop monitoring and management (Eastwood et al., 2017). This has substantially redefined significant parts of traditional farming practices, and therefore also the key competencies required in the sector. These disruptive innovations pose a significant challenge for agricultural incumbents, with external companies entering the sector with novel AI-based technologies developed in other contexts. IBM, through its Watson Decision Platform for agriculture, is a case in point, having made significant strides in predictive analytics for agriculture, raising the bar for traditional agricultural firms (Mostaco et al., 2018).

Our focus on the agricultural sector is poignant for several reasons. As farming is rooted in traditional practices, the integration of AI is a significant paradigm shift in the sector (Bronson and Knezevic, 2016). Moreover, as agriculture is a critical sector for global food security and sustainability, its possible responses to technological innovations could have wide-reaching implications (Misra et al., 2022). However, despite the growing interest in AI's potential to revolutionise agricultural practices, the precise mechanisms through which strategic renewal unfolds in this context remain under-researched. The interplay between incumbent companies such as DEERE & Co., KUBOTA KK, and AGCO Corp., and new entrants such as IBM, calls for deeper investigation into how these players adopt and adapt AI technologies.

3.2. Data

To examine the strategic renewal strategies of local and external companies in the agricultural playing field, we followed a five-stage approach.

First, we extracted patent data in alignment with the World Intellectual Property Organization's (WIPO) definition of AI-based inventions in the agricultural sector, which arose from a long-term collaboration between different intellectual property and AI specialists (WIPO, 2019a, b). In the literature on economics and management, patent data and company behaviour are widely recognised as suitable indicators of inventions and technological capability (Curran et al., 2010; Jaffe et al., 1993; Block et al., 2021; Berg et al., 2019; Borge et al., 2022; Waßenhoven et al., 2023). Our patent data on AI-based inventions in the agricultural sector was collected from the Derwent Innovation by means of search queries featuring combinations of keywords and patent classifications developed by the WIPO (see Appendix 1 (WIPO, 2019a, b)), which yielded 8483 patent families.

Second, we narrowed the focus to only the top 16 companies in this market, defined as those which filed at least 20 AI-based patent families in our chosen study period of 2000 to 2020. We took 2000 as our starting point because AI technologies have gradually evolved since then (WIPO, 2019b). To keep the focus on companies, we excluded universities from our sample, which was thus reduced to 813 unique AI-based agricultural patent families. Of these, 38 were co-filed by multiple applicants. To avoid bias, duplicates were counted only once in our descriptive analysis, but were considered for each of their co-applicants during regression analysis.

Third, we complemented our dataset with the full agricultural patent portfolios of the selected companies (14,412 unique patent families). Including non-AI patents allowed us to cluster each invention according to different types of AI-based inventions (cf. Table 1). Within this dataset, we focused on patents filed from 1990 onwards, although those up until 1999 were only used for control purposes (to derive prior patent stock). In total we extracted 7891 patent families (AI and non-AI) for the top 16 companies.

Fourth, we used the Standard Industrial Classification (SIC) codes to classify the selected companies into three industries (agricultural, machinery, and electronics).

Fifth, as we focused on one sector alone, we categorised the different

Table 1
AI-based agricultural inventions by companies from different industries.

Sector	Firms	AI patent families	Non-AI patent families	Total per sector (including duplicates)
Agriculture	6	390	4176	4566
Machinery	5	287	2314	2601
Electronics	5	162	600	762
Total (including duplicates)	16	839	7090	
Total (unique patent families)	16	813	7078	

AI-invention types using the most detailed level of the Cooperation Patent Classification (CPC), namely the 7-digit codes (Table 2), following others (e.g. Fleming, 2001). We classify AI patents as *Discontinuous Strategic Renewal Inventions (DSRI)* when a patent was filed in a new-to-the-company agricultural class (i.e., CPC-7 class in A01). *Follow-on DSRI* were patents filed in the class in which the company had already applied for a *DSRI*, while *Continuous Strategic Renewal Inventions (CSRI)* are patents filed in a class that was already familiar to the company. Non-AI patents were also examined to analyse the extent of AI integration with respect to the complete company portfolio.

3.3. Impact analysis of AI-based inventions

To answer our research questions, we assessed the impact of each AI-invention type on future technological developments by computing forward citations, a widely accepted proxy for technological impact and patent quality (Jaffe and de Rassenfosse, 2019). Higher citations counts have been linked both to the economic value and technological quality of inventions (Moser et al., 2018; Harhoff et al., 2003). As our study covers a long period (2000–2020), we normalised citation counts to address the bias caused by older patents receiving more citations. Specifically, we applied a yearly maximum standardisation as follows:

$$Y_i = \frac{C_{i,t}}{\text{Max}(C_t) + 1}$$

where Y_i is the standardised citation count, $C_{i,t}$ is the patents forward citation count, and $\text{Max}(C_t)$ is the highest citations count for year t . Y_i ranges between 0 and 1, with values closer to 1 for patents reaching near top values. While patents from recent years may receive fewer citations, this approach sufficiently distinguishes high from low impact patents. To test our assumptions on the effect of different renewal strategies, we ran an econometric model to assess the impacts on future inventions of each AI and non-AI invention for the 16 companies in the dataset.

First, we tested the impact of inventions based on their classification as AI-based, non-AI-based, and by sector (agriculture, electronics, and machinery). Second, we posit that an invention's impact is influenced by a company's specific characteristics, such as its prior stocks of inventions (total patents), agriculture-related inventions, AI-based inventions, and stocks for each invention type. We also consider the number of inventors involved and the technological scope of the invention (CPC scope counts), which are used as control variables in the econometric model. Third, we assess whether the impact of a specific AI type on future

Table 2
Operationalisation of AI-based invention typology.

AI-based invention type	Operationalisation via Cooperation Patent Classification (CPC) codes
Discontinuous Strategic Renewal Invention (DSRI)	AI patent in a company-new CPC-7 code A01
Follow-on DSRI	AI patent in the same CPC-7 code, where the company already applied for a DSRI
Continuous Strategic Renewal Invention (CSRI)	AI patent in a company-known CPC-7 code

technological developments varies across industries by considering interactions between AI type and sector. Detailed definitions of the variables are shown in Table 3. Fourth, we used negative binomial regression to estimate the probability of a count output variable (standardised forward citation count), where Y_i is the outcome of invention i , X_i is the AI type, I_i is the sector, and φ_i is a vector of control. We use the following regression equation to estimate the effects of invention type and industries on future technological developments:

$$Y_i = f(\beta X_i + \alpha I_i + \varphi_i + \varepsilon_i)$$

4. Findings

4.1. Descriptive statistics

The descriptive statistics of the 16 selected companies are presented in Table 4. These firms account for 9.9% of all AI-related patent families in the agricultural sector. Within the dataset, AI-related patent families represent 9% of all agricultural (i.e., local) company portfolios,

Table 3
Variable definitions.

Variable	Status	Definition
Standardised forward citation count	DV	Standardised count based on the yearly maximum of forward citations received by the patent family.
DSRI	IV	A dummy variable equal to 1 if the patent family is a discontinuous strategic renewal invention, otherwise 0.
Follow-on DSRI	IV	A dummy variable equal to 1 if the patent family is a follow-on discontinuous strategic renewal invention, otherwise 0.
CSRI	IV	A dummy variable equal to 1 if the patent family is a continuous strategic renewal invention, otherwise 0.
Non-AI	IV	A dummy variable equal to 1 if the patent family is a non-AI-based invention, otherwise 0.
Agriculture	IV	A dummy variable equal to 1 if the company that filed the patent family has an SIC code in the agricultural sector, otherwise 0.
Electronics	IV	A dummy variable equal to 1 if the company that filed the patent family has an SIC code in the electronics sector, otherwise 0.
Machinery	IV	A dummy variable equal to 1 if the company that filed the patent family has an SIC code in the machinery sector, otherwise 0.
Stock agriculture patents	Control	Number of patent families filed in the agricultural sector (CPC = A01*) by the company from 1990 to the $n - 1$ application year.
Stock AI patents	Control	Number of AI-based patent families filed by the company from 1990 to the $n - 1$ application year.
Stock all patents	Control	Number of patent families filed by the company from 1990 to the $n - 1$ application year.
Stock DSRI	Control	Number of discontinuous strategic renewal patent families filed by the company from 1990 to the $n - 1$ application year.
Stock Follow-on DSRI	Control	Number of follow-on discontinuous strategic renewal patent families filed by the company from 1990 to the $n - 1$ application year.
Stock CSRI	Control	Number of continuous strategic renewal patent families filed by the company from 1990 to the $n - 1$ application year.
Inventor count	Control	Number of distinct inventors who filed the patent family.
CPC scope count	Control	Number of CPC-7 codes in the patent family.

DV = dependent variable, IV = independent variable, DSRI = Discontinuous Strategic Renewal Invention, CSRI = Continuous Strategic Renewal Invention, CPC = Cooperative Patent Classification, SIC = Standard Industrial Classification.

Table 4

Descriptive statistics for the econometric model variables.

Variable	DSRI	Follow-on DSRI	CSRI	AI	Non-AI	All patents	Min.	Max.
	N = 331	N = 232	N = 275	N = 838	N = 7053	N = 7891		
	Mean			Mean				
Standardised forward citation count	0.059	0.053	0.034	0.049	0.045	0.05	0.000	0.966
Stock agriculture patents	622.930	1105.76	1159.59	932.71	1109.99	1091.16	0.0	3289.0
Stock AI patents	727.10	427.53	247.36	486.73	217.76	246.32	0.0	9658.0
Stock all patents	24,491.06	21,803.38	14,680.59	20,527.55	13,705.11	14,429.63	0.0	113,522.0
Stock DSRI	8.21	15.25	10.23	10.82	9.70	9.82	0.0	44.0
Stock Follow-on DSRI	4.05	9.78	6.53	6.45	6.75	6.72	0.0	35.0
Stock CSRI	5.07	11.97	22.17	12.59	10.35	10.59	0.0	64.0
CPC scope counts	9.82	9.19	8.65	9.26	4.73	5.21	1.0	69.0
Inventor counts	3.37	3.06	2.71	3.07	2.75	2.78	1.0	20.0
Standardised forward citation count × agriculture	0.074	0.052	0.034	0.049	0.049	0.049	0.966	0.000
Standardised forward citation count × Electronics	0.062	0.054	0.000	0.058	0.052	0.053	0.928	0.000
Standardised forward citation count × Machinery	0.045	0.054	0.039	0.046	0.038	0.039	0.931	0.000
Percentage								
Agriculture	2%	2%	4%	9%	91%	57%	0	1
Electronics	14%	7%	1%	21%	79%	10%	0	1
Machinery	5%	3%	3%	11%	89%	33%	0	1
At least one forward citation	30%	26%	22%	26%	30%	30%	0	1

DSRI = Discontinuous Strategic Renewal Invention, CSRI = Continuous Strategic Renewal Invention, CPC = Cooperative Patent Classification.

compared to 21% and 11% for electronics and machinery (i.e. external) companies, respectively (see Appendix 2, Table A2-1 for the list of companies, industries, and patent portfolios, Table A2-2 for the number of patent families per type, and Table A2-3 for variable standard deviations). External companies may not have agriculture as their primary Standard Industrial Classification (SIC) code, but can have an existing portfolio of largely non-AI-based patents in the agricultural playing field (79% and 89% for electronics and machinery companies, respectively).

Local and external companies differ in how they harness AI-related inventions for strategic renewal. Among local agricultural companies, just 2% of their AI patents are classed as *DSRI* while almost half can be characterised as *CSRI*. In contrast, external electronics companies file the highest share of agricultural *DSRI* (14%), and electronics companies have the lowest share of *CSRI* (1%). In other words, electronics are making major use of AI to break into new directions within agriculture. Machinery firms file 5% of all AI-related patent families classed as *DSRI*, with the remainder split almost equally between *Follow-on DSRI* and *CSRI*.

It would appear that agricultural incumbents mainly use AI to expand their existing portfolios via exploitative innovation, as they have the highest share of *CSRI*, while external machinery companies seem to enter the agricultural sector by expanding and improving their existing portfolios with AI (5% *DSRI*, 3% *CSRI*). External electronics companies mainly do so by implementing exploratory innovations, as reflected by their higher share of *DSRI* (14%), a trend that is also evolving over time (see Appendix, Fig. A2-1).

As for the long-term impact of these practices, which is measured using normalised forward citations, *DSRI* generally have the highest average influence on future inventions (0.059 standardised forward citations per patent family, i.e., reaches nearly 6% of the citation count of the top patent for its year). In turn, *Follow-on DSRI* have, on average, less impact than the first attempt (0.053 forward citations per patent family). Focusing on differences between local and external companies, we find that, on average, agricultural incumbents derive higher value from *DSRI* than machinery or electronics companies (0.074 vs. 0.062 and 0.045, respectively). Regarding incremental renewal strategies, local agricultural companies again outperform electronics companies (whose inventions received no forward citations), but are behind machinery companies. However, due to the differing profiles of these inventions and companies, a formal estimation model is needed to test these assumptions.

4.2. Model estimation

Our regression analyses confirm that the impact of AI-based strategic renewal on future technological development mainly depends on the type of invention that the company is developing. As shown in Table 5, AI-based *DSRI* have a higher impact on future technology developments than *CSRI*.

Our measurements of standardised forward citation counts underscore how firms pursuing *CSRI* experience a stronger negative impact on future inventions than those primarily engaging in *DSRI* (0.580 incidence rate ratio, Model 3). *DSRI* generates a new stream of knowledge outside of the firm's core innovation domain that future inventions can build on. Thus, more future inventions rely on knowledge produced by *DSRI* inventions than *CSRI*.

We confirmed these findings by performing robustness checks using other model specifications, as shown in Appendix 2, where Table A2-4 presents estimates using a binary dependent variable (i.e., at least one citation per invention), and Table A2-5 presents forward citations multiplied by 100 to account for potential limitations of using fractional numbers in negative binomial regression (Papke and Wooldridge, 1996, 2008).

Following earlier studies that have employed non-linear models with interaction terms (Buis, 2010), we interpret our results using predictive margins for each cluster by type of invention, which represent the average predicted standardised forward citation count. Table A2-1 in the Appendix shows these predictive margins for Model 3. By comparing the impacts of different renewal strategies, we find that AI-driven *DSRI* have a stronger impact on future inventions than *CSRI*, and the difference is statistically significant across all three industries in the agricultural playing field (see Fig. 1 and Table A2-3). Indeed, *DSRI* helps companies to subsequently seize new opportunities by investing in further inventions, and this *Follow-on DSRI* tends to have a significantly even higher impact than earlier *DSRI* attempts. This diverges from our descriptive findings, where initial *DSRI* received the highest standardised citation count. It would seem that once companies introduce *DSRI* and enter new business fields, they tend to gain greater returns on their cumulative knowledge by following a discontinuous renewal strategy.

We then restricted our analysis to each sector separately. Our findings for machinery companies require partial correction because these firms receive a high number of citations of their *DSRI* and *CSRI* (Fig. 2). Our previous results are also only partially supported with regard to electronics and agricultural companies because the difference between

Table 5
Impact model estimation of inventions.

Dependent variable:	Standardised forward citation count	Standardised forward citation count	Standardised forward citation count	Standardised forward citation count
Model:	(1) Neg. binomial	(2) Neg. binomial	(3) Neg. binomial	(4) Linear
Follow-on DSRI	0.897 (0.142)	1.307** (0.166)	1.079*** (0.0258)	−0.00299* (0.000786)
CSRI	0.580*** (0.116)	0.721 (0.158)	0.580*** (0.0260)	−0.0301*** (0.000320)
Non-AI	0.769*** (0.0641)	0.894 (0.0741)	0.779*** (0.0370)	−0.0170*** (0.00144)
Electronics	1.043*** (0.0124)	0.651*** (0.106)	0.573*** (0.0493)	−0.0321** (0.00352)
Machinery	0.799*** (0.00230)	0.729** (0.108)	0.541*** (0.0482)	−0.0374*** (0.00373)
Stock agriculture patents		1.000* (0.0000789)	1.000* (0.0000845)	−0.00000834* (0.00000274)
Stock AI patents		1.000 (0.0000833)	1.000 (0.0000902)	0.00000604 (0.00000312)
Stock all patents		1.000 (0.00000389)	1.000 (0.00000413)	−0.000000177 (0.000000132)
Stock DSRI		0.973* (0.0142)	0.973* (0.0142)	−0.00100 (0.000396)
Stock Follow-on DSRI		0.980*** (0.00329)	0.980*** (0.00324)	−0.000205 (0.000257)
Stock CSRI		0.987*** (0.00453)	0.987*** (0.00440)	−0.000533*** (0.0000372)
Inventor counts		0.989 (0.0204)	0.990 (0.0206)	−0.000333 (0.000861)
CPC scope count		1.016*** (0.00170)	1.016*** (0.00144)	0.000706*** (0.0000321)
Follow-on DSRI × Electronics			1.032 (0.124)	0.00500 (0.00421)
Follow-on DSRI × Machinery			1.727*** (0.229)	0.0258*** (0.00189)
CSRI × Electronics			0.000000230*** (0.000000287)	−0.0267** (0.00407)
CSRI × Machinery			1.924*** (0.125)	0.0372*** (0.00157)
Non-AI × Electronics			1.170 (0.119)	0.0124* (0.00320)
Non-AI × Machinery			1.344*** (0.0945)	0.0191*** (0.00163)
Observations	7891	7891	7891	7891
Pseudo R ²	0.002	0.037	0.037	0.0398

DSRI = Discontinuous Strategic Renewal Invention, CSRI = Continuous Strategic Renewal Invention, CPC = Cooperative Patent Classification.

Notes: Robust standard errors in parentheses, clustered by sector. Results for negative binomial regressions are exponentiated coefficients (odds ratios). The base project type is agriculture, discontinuous strategic renewal invention.

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

their first attempt at *DSRI* and their *CSRI* may be positive, but is not statistically significant at the 95% confidence interval (CI).

We then compared differences between the impacts of each strategy across local and external companies in the playing field. Here, we found that the *CSRI* strategies of agricultural companies have a more substantial impact on future inventions than those of electronics companies (whose *CSRI* received no citations). The impact is similar – but not significantly different – from that of machinery firms. Thus, our findings are only supported in terms of the impact of local agricultural companies' *CSRI* as opposed to that of their external electronics counterparts. Local companies' *DSRI* strategies also have a more substantial impact on future inventions than those of external companies, suggesting that agricultural companies derive greater value from integrating AI technologies into new-to-company applications than when electronics and machinery companies do so when pursuing the same strategic approach. Our findings are summarised in Table 6.

5. Discussion

Our study investigates how incumbent companies and new market entrants implement AI in their innovation portfolios by introducing a new typology of AI technology usage. *Discontinuous Strategic Renewal Inventions (DSRI)* are those that create new-to-company applications that open new technological paths. *Follow-on DSRI* showcase incumbent's efforts to further develop these paths, while *Continuous Strategic Renewal Inventions (CSRI)* extend the existing innovation portfolio without disrupting current business models (regarding existing technological paths). We then use this typology to explore the use of AI in new technology development, basing our methodology on patent analysis. We were able to measure which types of AI have the greatest impact on future technological developments by examining forward citations, a common proxy for the capacity of companies to enter and thrive in new markets. Our evaluation of these effects by company profiles (i.e., local and external) have provided novel insights into how companies engage in strategic renewal and refresh their technological capabilities, in this case focusing on the agricultural playing field.

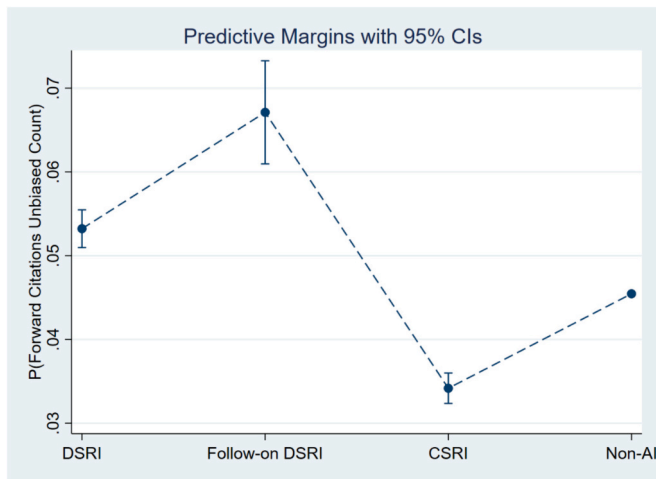


Fig. 1. Probability of outputs across invention types. Note: Regression data for the plots are shown in Table 5 — Model 3.

First, we show that *DSRI* strategies have a greater impact on future technological developments than *CSRI* do, but *Follow-on DSRI* have the greatest impact of all. Second, technology-based strategic renewal is contingent on the company's primary sector. Predominantly agricultural companies derive greater benefit than other players from outside this playing field from integrating AI technologies into new-to-company applications.

5.1. Implications for strategic renewal and the AI literature

Our work lies at the crossroads of strategic renewal and innovation management, specifically in the context of AI implementation. We make especially pertinent contributions to the literature on technology-based strategic renewal by developing a new typology of AI-based inventions, which helps to understand how incumbents implement new technologies, open up new-to-company businesses and support 'avianisation', i.e. their capacity to strategically renew their resources to remain competitive (Roy et al., 2018).

We also contribute to this stream of literature by demonstrating that the value derived by companies from their pursuit of both discontinuous

and incremental renewal strategies is contingent on their core industries, and particularly that local companies engaging in the former achieve a more substantial impact on future inventions than external companies do, contradicting the prevailing assumptions in the literature (Agarwal and Helfat, 2009; Thrane et al., 2010; Blomkvist et al., 2010). Local companies may be better positioned to exploit their existing market expertise in their own fields and thus reduce the 'burden of knowledge' (Arts and Fleming, 2018) when engaging in discontinuous renewal. As digital technologies complement firms' existing knowledge stocks (Lanzolla et al., 2021; Bohnsack et al., 2021; Bohnsack et al., 2024), the benefits tend to be greater for local companies due to their established expertise in the playing field, while their external counterparts are less equipped to provide relevant AI-based inventions in an unfamiliar setting (cf. Bohnsack et al., 2024). Third, we demonstrate that firms must invest in *Follow-up DSRI* if they want to maximise impact on future technological developments. Indeed, initial attempts at *DSRI* in the agricultural playing field yield lower returns than continued investment in the same explorative paths. Finally, we find that AI-based inventions unlock emerging market opportunities for both incumbents and new entrants in the agricultural playing field, leading to the kind of interdependencies that have been described in previous literature (Agarwal and Helfat, 2009; Bergek et al., 2013; Schmitt et al., 2018), whereby local companies integrate emerging knowledge with their existing expertise in order to stay competitive and keep pace with external companies diversifying the market (cf. Bohnsack et al., 2024). Thus, our findings support the concept of creative accumulation, i.e., the improvement of existing technologies through the integration of new knowledge (Bergek et al., 2013; Pavitt, 1986).

Our new typology also contributes to the literature on AI technology in management science (Ceipek et al., 2021a; Ceipek et al., 2021b; Plantec et al., 2023), which has typically focused on the diffusion of AI technologies and its impact on value chains in different markets (Tong et al., 2021; Gregory et al., 2021; Bradley and O'Toole, 2016), and on how AI can help innovators access and process information differently (Lanzolla et al., 2021; Marion and Fixson, 2021; Verganti et al., 2020; Kroh et al., 2018; Sestino et al., 2020). We have added to this stream by focusing explicitly on the different ways in which companies implement AI technologies in their innovation portfolios, building on earlier insights regarding other digital technologies (Ceipek et al., 2021a; Cappa et al., 2021). Cappa et al. (2021), for instance, noted the lack of indicators to assess the long-term impact of big data technologies on firm

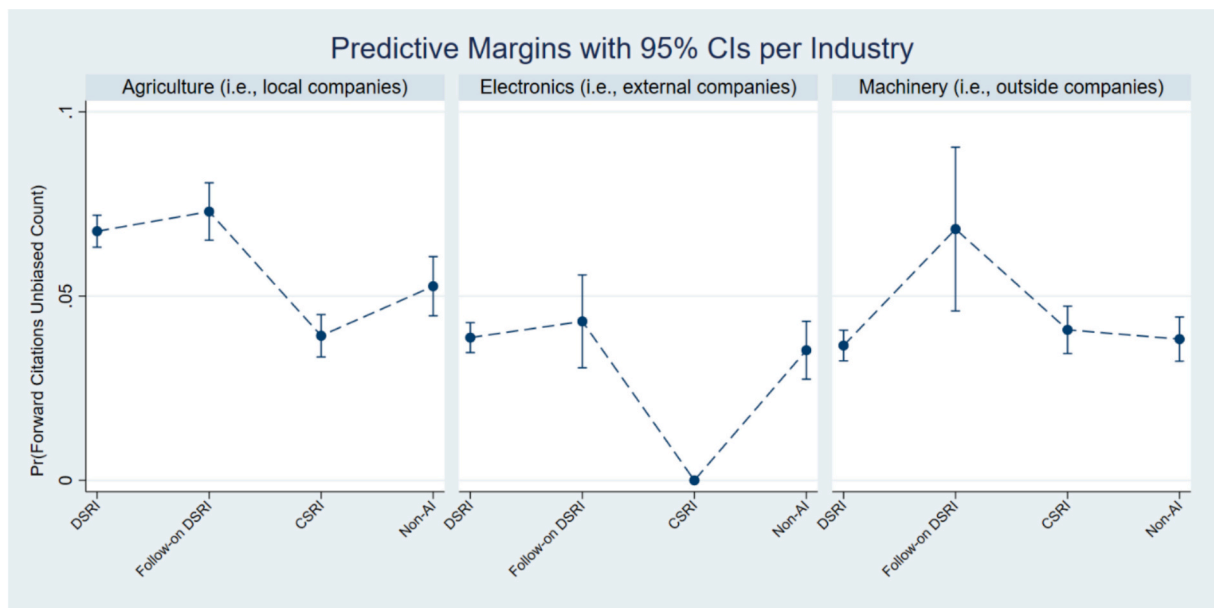


Fig. 2. Probability of outputs across invention types and industries. Note: Regression data for the plots are shown in Table 5 — Model 2.

Table 6

Summary of results. Note: for limitations see Figs. 1 and 2.

Categories	Findings	Limitations
Comparison of renewal strategies' performances	AI-based <i>DSRI</i> have a stronger impact on future inventions than <i>CSRI</i>	No limitation
	<i>Follow-on DSRI</i> have a stronger impact on future inventions than first-time <i>DSRI</i> .	No limitation
Comparison of renewal strategies' performances across local and external companies	<i>DSRI</i> of local companies have a stronger impact on future inventions than <i>DSRI</i> of external companies.	No limitation
	<i>Follow-on DSRI</i> of local companies have a stronger impact on future inventions than <i>Follow-on DSRI</i> of external companies.	Significant for agricultural (i.e., local) companies and electronics (i.e., external), but not statistically significant for machinery (i.e., external) companies (at 95% confidence intervals).
	<i>CSRI</i> of local companies have a stronger impact on future inventions than <i>CSRI</i> of external companies.	Significant for agricultural (i.e., local) companies and electronics (i.e., external), but not statistically significant for machinery (i.e., external) companies (at 95% confidence intervals).

DSRI = Discontinuous Strategic Renewal Invention, *CSRI* = Continuous Strategic Renewal Invention.

performance, while proposing that implementation costs may outweigh the benefits. Our study complements these findings by presenting a patent-based method to analyse the impact of AI on future technological developments, and showing that the latter is contingent on the company's original playing field. Our study also echoes [Ceipek et al.'s \(2021a\)](#) finding (for family-owned businesses developing IoT technologies) that incumbents vary in their commitment to explorative versus exploitative digital technology pathways, with the former being more closely tied to the diversification of previous knowledge and specific companies' managerial characteristics. In the case of large companies in the agricultural machinery sector, our findings align with [Ceipek et al. \(2021a\)](#) as we demonstrate that external companies more frequently pursue *DSRI* when it comes to AI technologies. However, we also find that local companies are nevertheless more likely to derive greater value from such innovations.

A possible explanation for our finding that less discontinuous approaches to AI integration in product innovation result in lower technological impact can be found in [Wang \(2021\)](#), who showed that such an exploitative usage of digital technologies, such as redesigning existing products, requires the adoption of specific innovation methodologies, such as design thinking.

5.2. Implications for practice

Our findings carry three important lessons for practitioners. First, the potential impact of different renewal strategies (disruptive or incremental) on future technological developments seems to be contingent on whether a company is local or external to a given playing field. Incumbents' specialised knowledge clearly supports the development of more targeted and impactful innovations and products. Second, regardless of origin, for all companies perseverance is a virtue when it comes to further developments of pioneering initiatives, in line with a discontinuous renewal strategy. Both local and external firms that are

willing to take risks are more likely to reap the rewards from more qualitative and valuable inventions. Third, we provide practitioners with a comprehensive classification based on patent data to assess their internal technology portfolios and monitor behaviour in their own markets and beyond. We also believe that our classification is equally applicable to other (emerging) technologies, and not just AI.

Our differentiation is consistent with broader societal discussions regarding the beneficiaries of AI innovation. Research has shown that AI adoption often amplifies existing inequalities by disproportionately benefiting high-income workers and regions with advanced digital infrastructure, while marginalised communities face barriers to access and use ([Domini et al., 2022](#)). However, this observation needs to be treated with caution, as innovation, when measured by patent applications, tends to be concentrated in urban and developed areas ([OECD, 2016](#)). This pattern is especially evident for AI-related inventions. For example, in Europe, 40% of all AI patent applications originate from just 23 of its 230 regions ([Buarque et al., 2020](#)), which clearly favour AI innovation due to their higher levels of investment in R&D, the presence of leading universities, research institutions and innovation hubs, and a greater concentration of skilled professionals in technology and AI-related fields. Local companies in the agricultural playing field may derive greater benefits from AI due to their proximity to sector-specific needs and infrastructure. This parallels the observation that local populations often enjoy better access to technology and social systems ([Ekoh et al., 2023](#)).

In agriculture, AI applications are diverse, ranging from yield prediction and precision farming to crop health monitoring and livestock management. For instance, AI-driven yield prediction models can help farmers make informed decisions about crop planting and resource allocation based on historical data, weather conditions, and soil properties. In precision farming, AI-powered drones can monitor field conditions, enabling farmers to optimize irrigation, fertilization, and pest control strategies — both underscore the need for human-centred AI approaches in agriculture (see e.g. [Holzinger et al., 2024](#); [Holzinger et al., 2022](#)). Hence, AI has the potential to revolutionise the agricultural sector, creating new market opportunities and competitive advantages for both incumbents and new entrants. By strategically integrating AI technologies into their operations, companies such as DEERE & Co. and IBM illustrate the benefits of embracing digital transformation in agriculture. Our findings highlight the need for those companies to not only take initial steps toward strategic renewal, but also to persevere in such endeavours.

5.3. Limitations and future research

While our empirical investigation has provided many useful insights, it is not without its limitations. For instance, our analysis is restricted to patent data within the agricultural playing field. Future research could include other industries and incorporate a wider range of company data beyond patents to assess the broader impact of strategic renewal practices, and not only on technological capabilities but also corporate performance and overall market development. Moreover, we also only focus on technology-driven strategic renewal, while other forms, such as new business models, are not reflected in patent analysis.

Acquisition of our data from a single outlet may introduce hidden biases, especially if it solely reflects the resource or technology-supply side, in the absence of a market or demand side. However, our analysis does observe strategic renewal over a prolonged period, which is important when comparing between discontinuous and incremental strategies, and is thus a useful complement to other qualitative research.

Our study offers a valuable overview of the general situation, but future case study-based research within specific companies challenged by major disruptions could help to further our understanding of how companies renew by developing their technology portfolios. Furthermore, a game theory perspective could enhance understanding of strategic decision-making in competitive markets. Investigating how firms

anticipate and react when rivals adopt AI-driven innovations could provide further insights into how incumbents' early investments in *DSRI* influence entrants' strategies or lead to new competitive equilibria.

Future research could also examine the knowledge sources behind AI implementation, perhaps using patents' backward citations to distinguish between internal and external knowledge, or exploring non-patent literature citations to identify technological or scientific foundations. Finally, as AI is often considered an enabling or general-purpose technology with wide-ranging applications, our typology could easily be applied to other industries besides agriculture, sparking conversations on strategic renewal and the transformation of innovation strategies associated with AI and other emerging technologies across diverse markets.

CRediT authorship contribution statement

Michael Rennings: Writing – review & editing, Writing – original draft, Validation, Project administration, Funding acquisition, Conceptualization. **Quentin Plantec:** Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Formal analysis, Conceptualization. **Anna Waßenhoven:** Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Formal analysis, Conceptualization. **Clement Sternberger:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis. **Mei Yun Lai:** Writing – review & editing, Conceptualization. **Stefanie Bröring:** Writing – review & editing, Resources, Funding acquisition.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT 4.0 and DeepL in order to ensure language editing. After using these tools/services, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.techfore.2026.124640>.

Data availability

Data will be made available on request.

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